

The argument from poverty of the stimulus

Computational modeling of language
Spring 2026

Bureaucracy

- Midterm grades coming up.

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- Spring break next week!

This week's readings

- Chomsky, *Knowledge of language*
- Foraker et al. on anaphoric *one*
- Chater & Vitanyi on simplicity
- Pullum on Chomsky 2011 in London

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Two problems of human knowledge

- **Plato's problem:** Why do we know so much, given so little evidence?
- **Orwell's problem:** Why do we know so little, given so much evidence?

Two problems of human knowledge

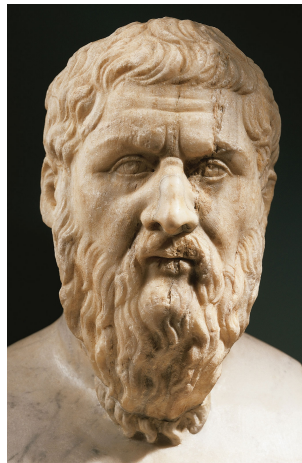
- **Plato's problem:** Why do we know so much, given so little evidence? **Poverty of the stimulus**
- **Orwell's problem:** Why do we know so little, given so much evidence?

The argument from poverty of the stimulus



Chomsky

The argument from poverty of the stimulus



Plato

Leibniz

Chomsky

The argument from poverty of the stimulus

- Children know things about language that they **could not have learned** from the input they receive.
- Therefore, at least some of this linguistic knowledge must be **innate**.

John is too stubborn to talk to Bill (6)

John is too stubborn to talk to (7)

Examples 6 and 7 are superficially analogous -
but only superficially.

John is too stubborn to talk to Bill	(6)
John is too stubborn to talk to	(7)

Examples 6 and 7 are superficially analogous - but only superficially.

How does a child learn the underlying **structure**?

The bear in the cave is hungry.

S: The bear in the cave is hungry.

Q: Is the bear in the cave hungry?

Question formation

Data:

S: The bear in the cave is hungry.

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Two hypotheses:

Question formation

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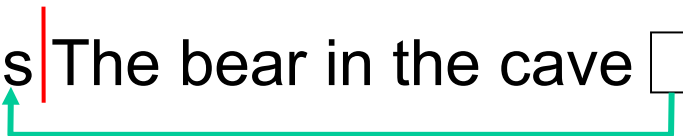
S: The bear in the cave is hungry.

Q: Is the bear in the cave hungry?

Two hypotheses:

Sequential: Front the first occurrence of “is”:

Is | The bear in the cave ☐ hungry?

A diagram illustrating the process of question formation. A red vertical line is placed after the word 'Is'. A teal line starts from the bottom of this red line, extends horizontally to the right, and then turns vertically upwards to point at the word 'is' in the sentence 'The bear in the cave ☐ hungry?'. This indicates that the 'is' from the original sentence is being moved to the front.

Question formation

Data:

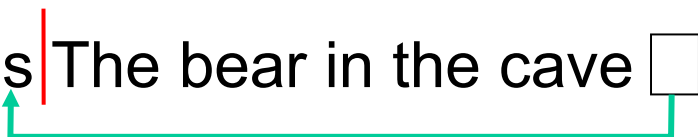
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Structure-dependent: Front the “is” that agrees with the subject NP of the sentence:

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Simpler!

Is | The bear in the cave ☐ hungry?



Structure-dependent: Front the “is” that agrees with the subject NP of the sentence:

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Question formation

More data:

S: The bear that is in the cave is hungry.

Q:

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Disambiguating data

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Question formation

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Disambiguating data
- but **absent** from
children's input!

Two hypotheses:

Sequential: Front the first occurrence of “is”:

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Is | ☐ The bear that is in the cave ☐ hungry?



Question formation

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Poverty of
the stimulus



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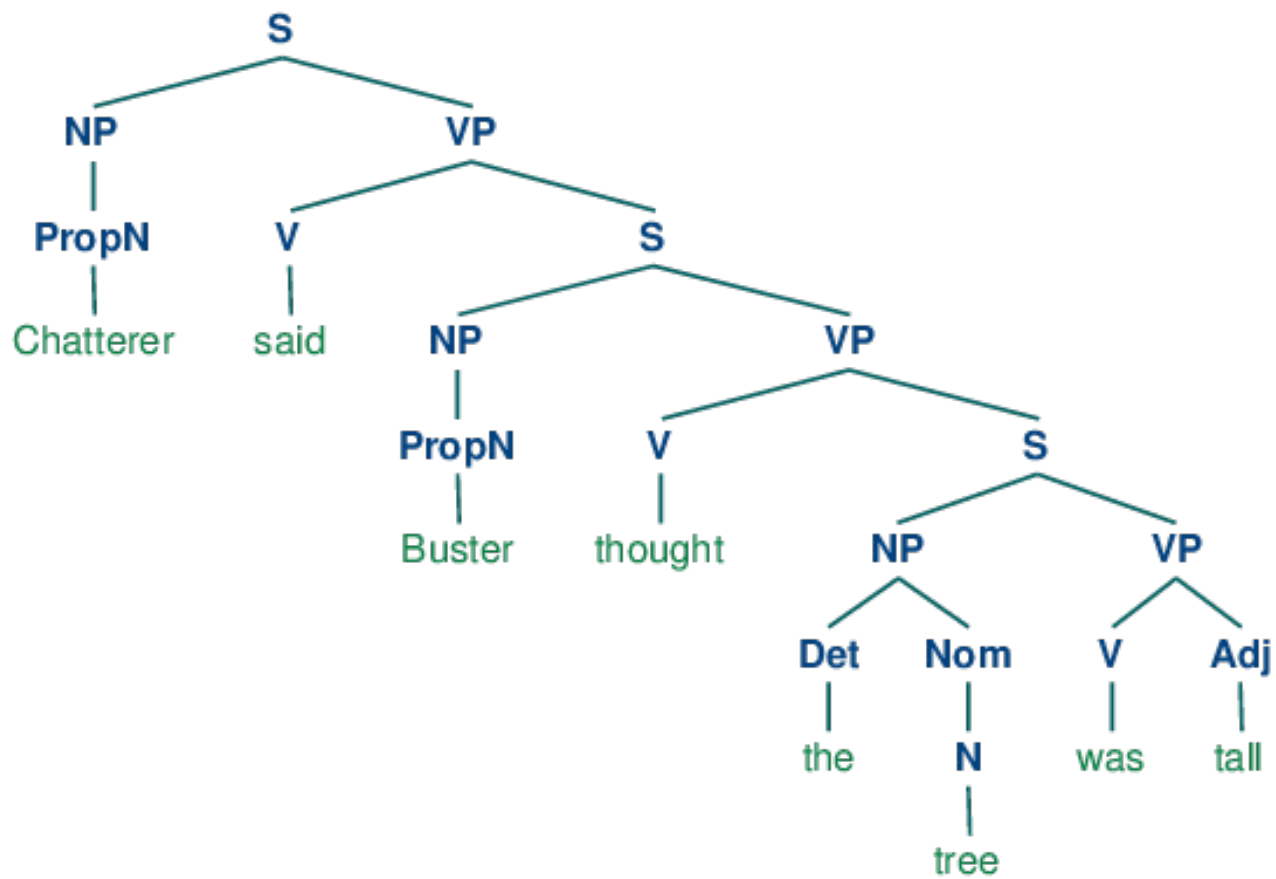


Yet children know!

3-to-5 year old children **correctly** form questions from declarative sentences in English.

Thus, young children **know the correct hypothesis**.

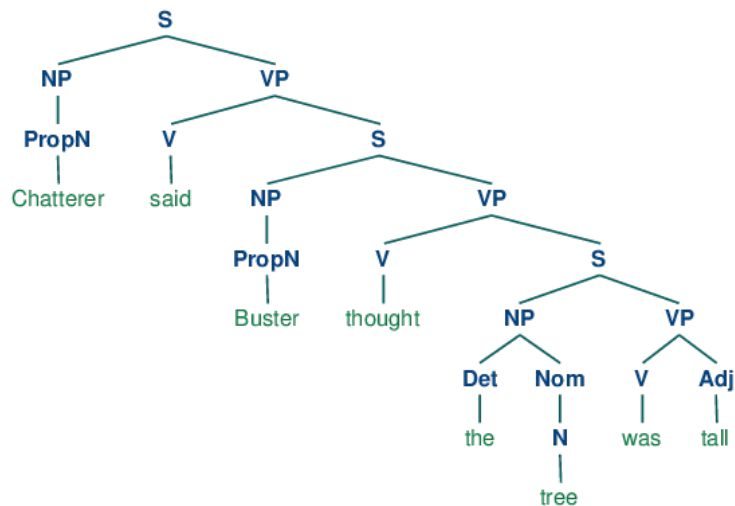
Crain & Nakayama, 1987



In sum:

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1. Children **know** that language is **hierarchically structured**.



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2. They **could not have learned** that knowledge from evidence alone.

**Poverty of
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3. Therefore, knowledge of language must be **at least partially innate**.

In sum:

1. Children **know** that language is **hierarchically structured**.
2. They **could not have learned** that knowledge from evidence alone.
3. Therefore, knowledge of language must be **at least partially innate**.

Small groups: Are you **convinced** by this argument?

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Indirect Evidence and the Poverty of the Stimulus: The Case of Anaphoric *One*

Stephani Foraker,^a Terry Regier,^a Naveen Khetarpal,^a
Amy Perfors,^b Joshua Tenenbaum^b

Core idea:

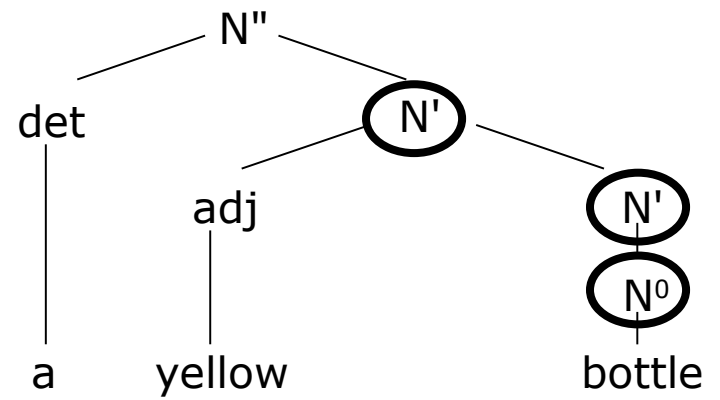
1. One can learn by noticing what is **systematically absent** from the stimulus.

Core idea:

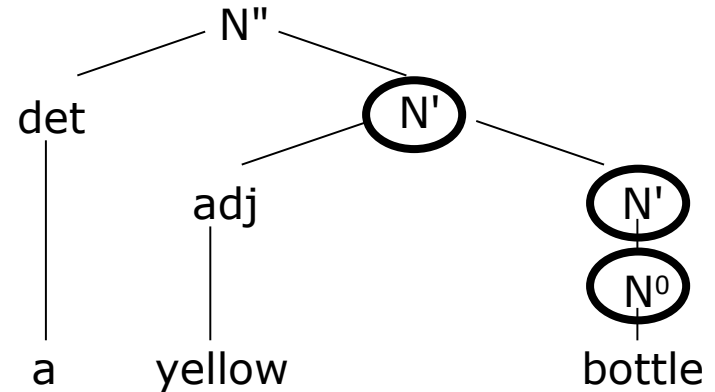
1. One can learn by noticing what is **systematically absent** from the stimulus.
2. Captured in Bayesian terms, specifically by the **likelihood** term $p(d|H)$.

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1. One can learn by noticing what is **systematically absent** from the stimulus.
2. Captured in Bayesian terms, specifically by the **likelihood** term $p(d|H)$.
3. Applied to a classic poverty-of-stimulus example, **anaphoric one** in English.

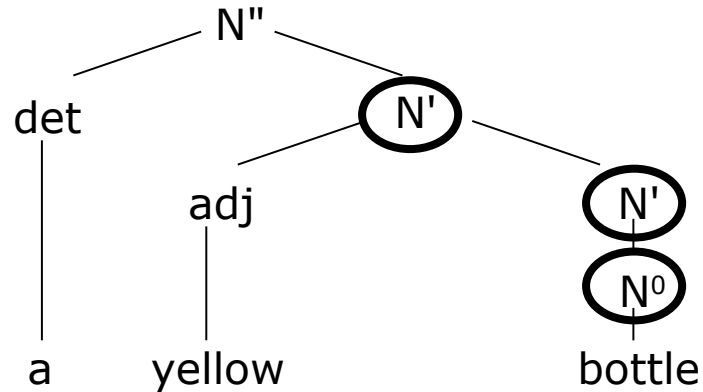


An established example of the argument from poverty of the stimulus.



An established example of the
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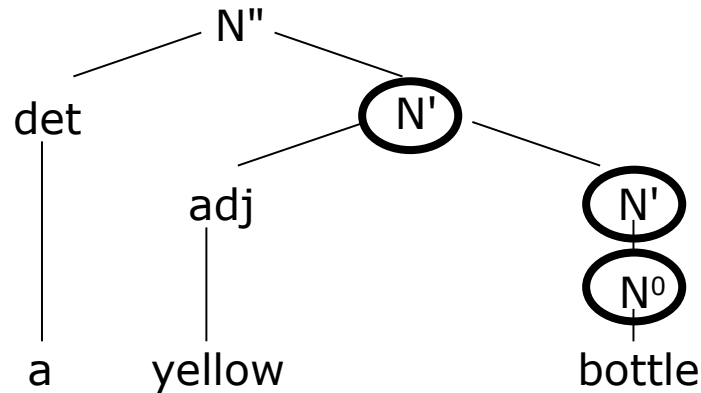
Here's a yellow bottle.
Do you see another **one**?



An established example of the
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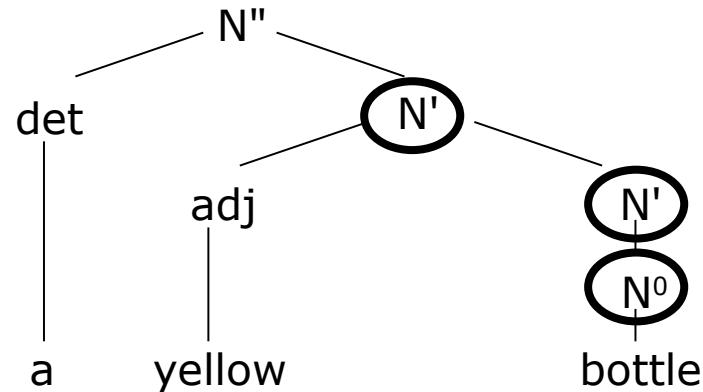


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Do you see another **one**?

Baker, 1978; Hornstein & Lightfoot, 1981;
Lidz, Waxman, & Friedman, 2003



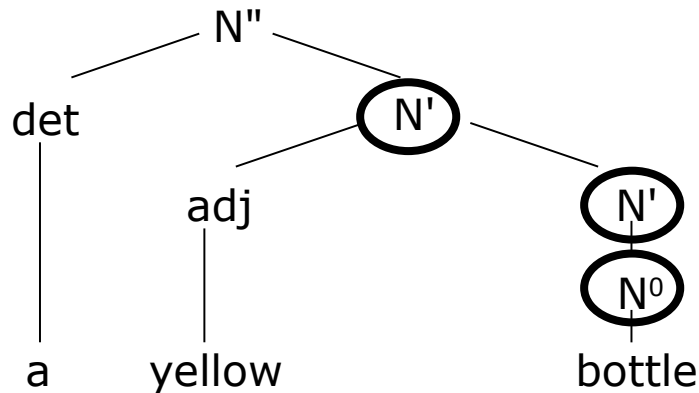
Challenge: Learn the antecedent of anaphoric *one* in a hierarchical NP.

An established example of the argument from poverty of the stimulus.

Here's a yellow bottle.

Do you see another ***one***?

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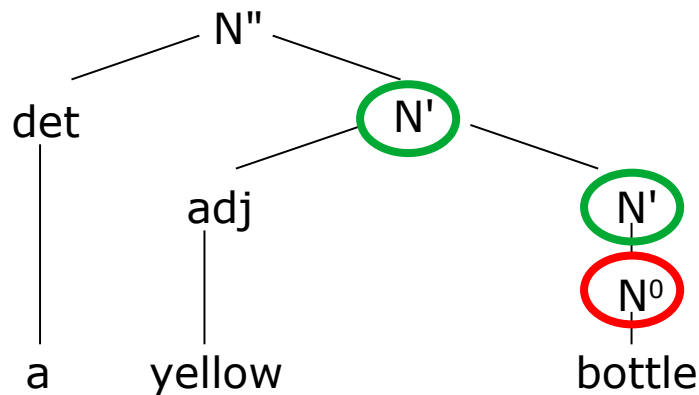


Correct hypothesis: *one* = any N'

Incorrect hypothesis: *one* = N⁰

Here's a yellow bottle.

Do you see another ***one***?



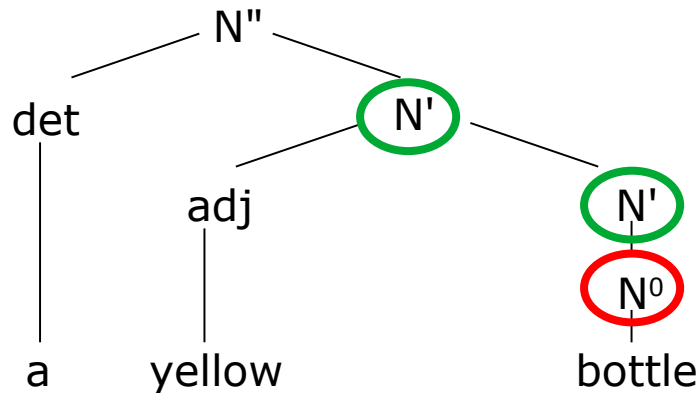
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Problem: Data that rule out N⁰ in one shot
are not available in input. [Lidz et al., 2003](#)

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Incorrect hypothesis: *one* = N^0

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Nativist conclusion: N^0 must be **innately excluded** during learning. Universal grammar.

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Nativist conclusion: N^0 must be **innately excluded** during learning. Universal grammar.

Will show: can learn from **realistic input** *without* innately excluding N^0 . **How?**

Maybe the child learns about *one* the same way a linguist does:

by noticing the different behavior of **complements** and **modifiers**.

Complements vs. modifiers

Syntactic distinction, conceptual basis.

A **complement** is evoked conceptually by its head noun, but a **modifier** is not.

*member **of congress***

*ball **with stripes***

Complements vs. modifiers

*I want the **ball**
with stripes.*

*You can have the **one**
with dots.*

Complements vs. modifiers

*I'll have the **piece**
of cheese.*

*I want the **ball**
with stripes.*

*You can have the **one**
with dots.*

Complements vs. modifiers

*I'll have the **piece**
of cheese.*

You can have the **one
of apple.*

*I want the **ball**
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Complements vs. modifiers

'One' **cannot** index a complement-taking noun without its complement.

*I'll have the **piece** of cheese.*

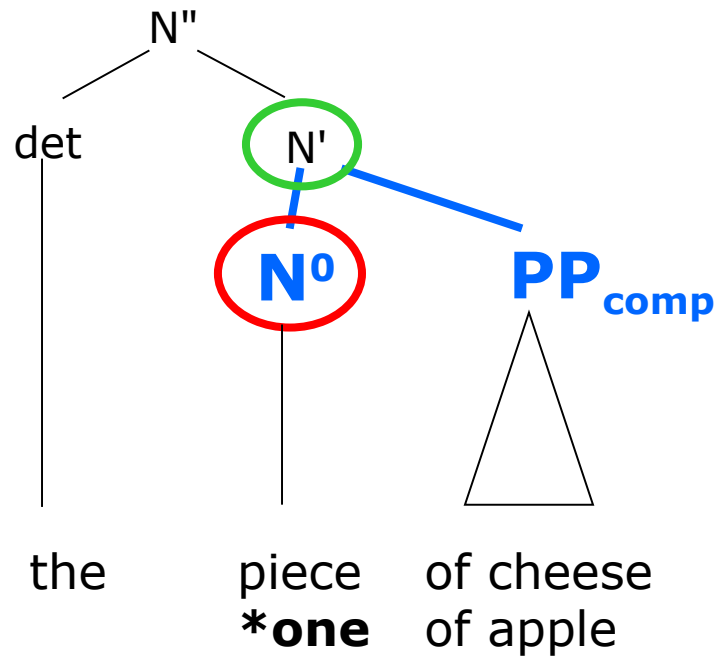
You can have the **one of apple.*

'One' **can** index a noun without its modifier.

*I want the **ball** with stripes.*

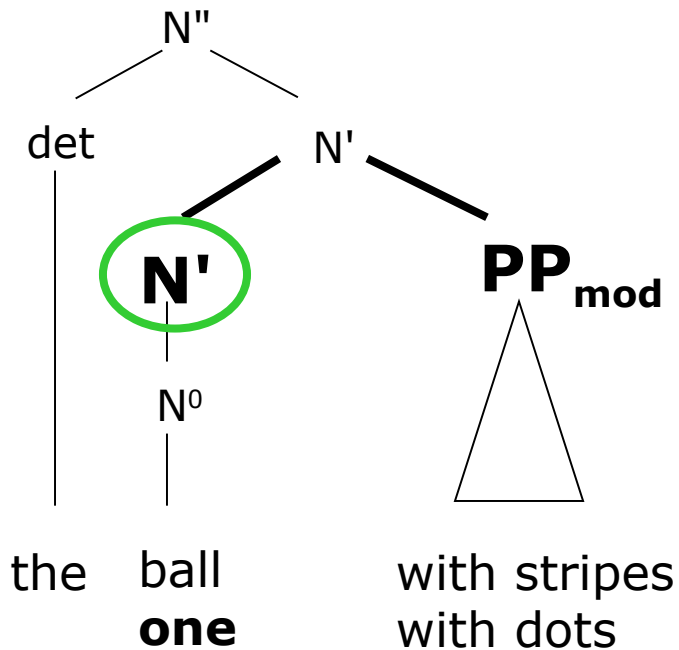
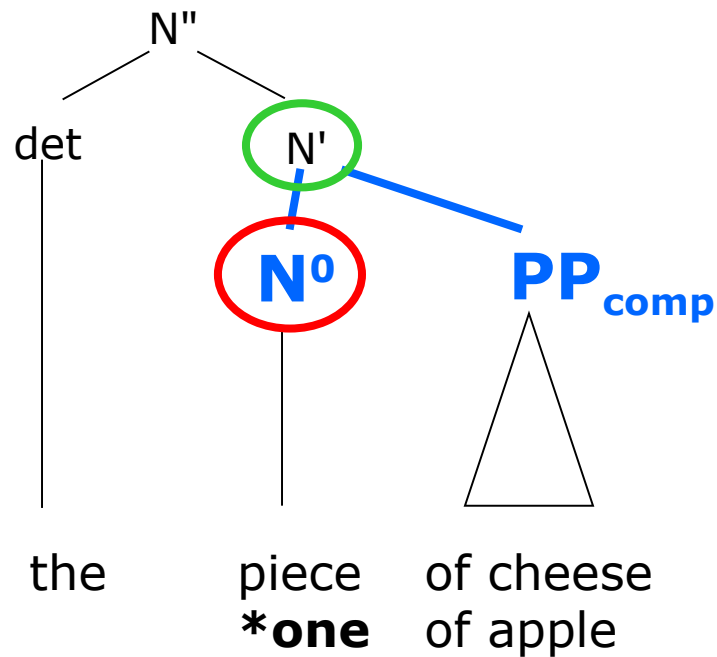
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Complements vs. modifiers



Jackendoff, 1977; Lidz & Waxman, 2004; Radford, 1998

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Given realistic input,

is there enough evidence for a child
to **learn** this?

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Lidz et al. 2003: **NO**

We claim: **YES**

Data

Noun phrases

- Nina corpus (Suppes, 1974) in CHILDES, 23-29 mos.
60,000+ noun phrases, from 34,300 child-directed utterances.

Selected random cumulative samples

- 5%, 10%, 15%
- 4.2 hrs, 8.4 hrs, 12.6 hrs

Data

Each NP coded as a sequence of part-of-speech tags, with special tags for complements and modifiers.

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the green monkey = determiner adjective noun

a piece of cheese = determiner noun **complement**

the ball with stripes = determiner noun **modifier**

him = pronoun

that one = determiner anaphoric-one

the one by your foot = determiner anaphoric-one **modifier**

Hypotheses

one = any N'

one = N^0

Hypotheses

one = any N'

Det → determiner

Adj → adjective

PossPro → possessive pronoun

AposS → apostrophe s

Mod → modifier

Comp → complement

Nzero → noun

Pro → pronoun

NP → Pro | Nbar | Det Nbar | Poss Nbar

Poss → NP AposS | PossPro

Nbar → Poss Nbar | Adj Nbar | Nbar Mod

Nbar → Nzero | Nzero Comp

Nbar → anaphoric one

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Nzero → anaphoric one

≠

Scoring

$$p(H|D) \propto p(D|H) p(H)$$

Scoring

$$\boxed{p(H|D)} \propto \boxed{p(D|H)} \boxed{p(H)}$$

posterior likelihood prior

Scoring

Score Fit to data Simplicity

↓ ↓ ↙

$$\boxed{p(H|D)} \propto \boxed{p(D|H)} \boxed{p(H)}$$

posterior likelihood prior

The diagram illustrates Bayes' theorem with three annotations. 'Score' points to the posterior $p(H|D)$. 'Fit to data' points to the likelihood $p(D|H)$. 'Simplicity' points to the prior $p(H)$. Each term in the equation is enclosed in a blue box, and the labels below are also in blue.

Scoring

Score Fit to data Simplicity

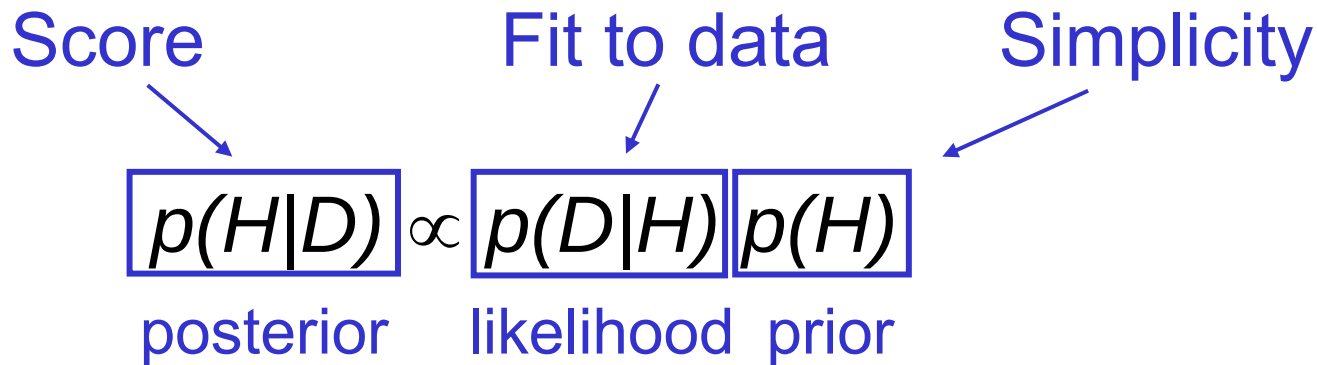
↓ ↓ ↙

$$\boxed{p(H|D)} \propto \boxed{p(D|H)} \boxed{p(H)}$$

posterior likelihood prior

The 2 grammars are equally complex, so
 $p(N') = p(N^0)$.

Scoring



The diagram shows the equation $p(H|D) \propto p(D|H)p(H)$ with three blue arrows pointing to its components. The arrow from 'Score' points to $p(H|D)$, the arrow from 'Fit to data' points to $p(D|H)$, and the arrow from 'Simplicity' points to $p(H)$. Below each term is its name: 'posterior' for $p(H|D)$, 'likelihood' for $p(D|H)$, and 'prior' for $p(H)$.

$$\boxed{p(H|D)} \propto \boxed{p(D|H)} \boxed{p(H)}$$

Score Fit to data Simplicity

posterior likelihood prior

All learning is driven by the likelihood.

Likelihood

Adjusted production probabilities in each grammar to maximize likelihood $p(\text{corpus}|\text{grammar})$, via inside-outside algorithm.

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Both grammars are **consistent** with input.

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Difference:

N^0 grammar **also** predicts the form:

[*one* complement] (e.g. *one of cheese*).

N' grammar **does not**.

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“Wasted” probability mass!

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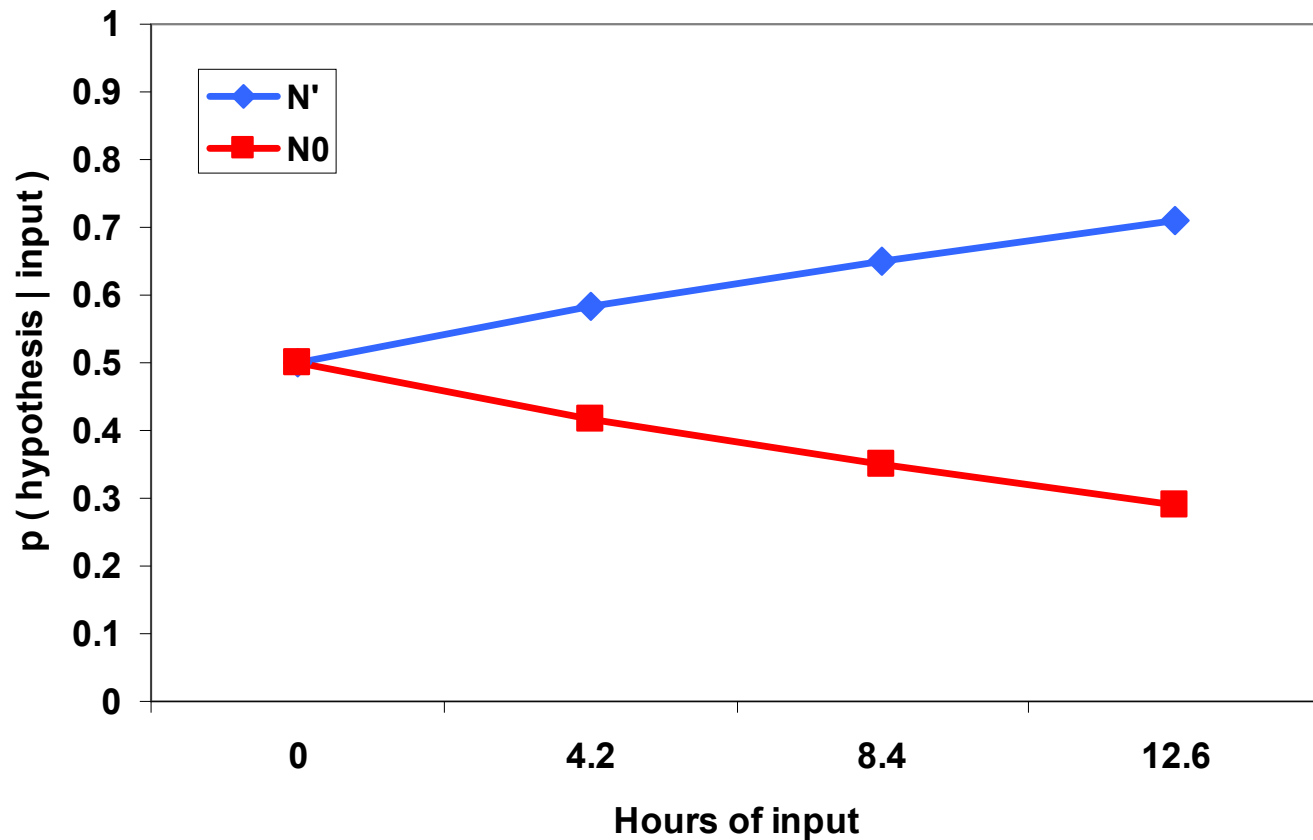
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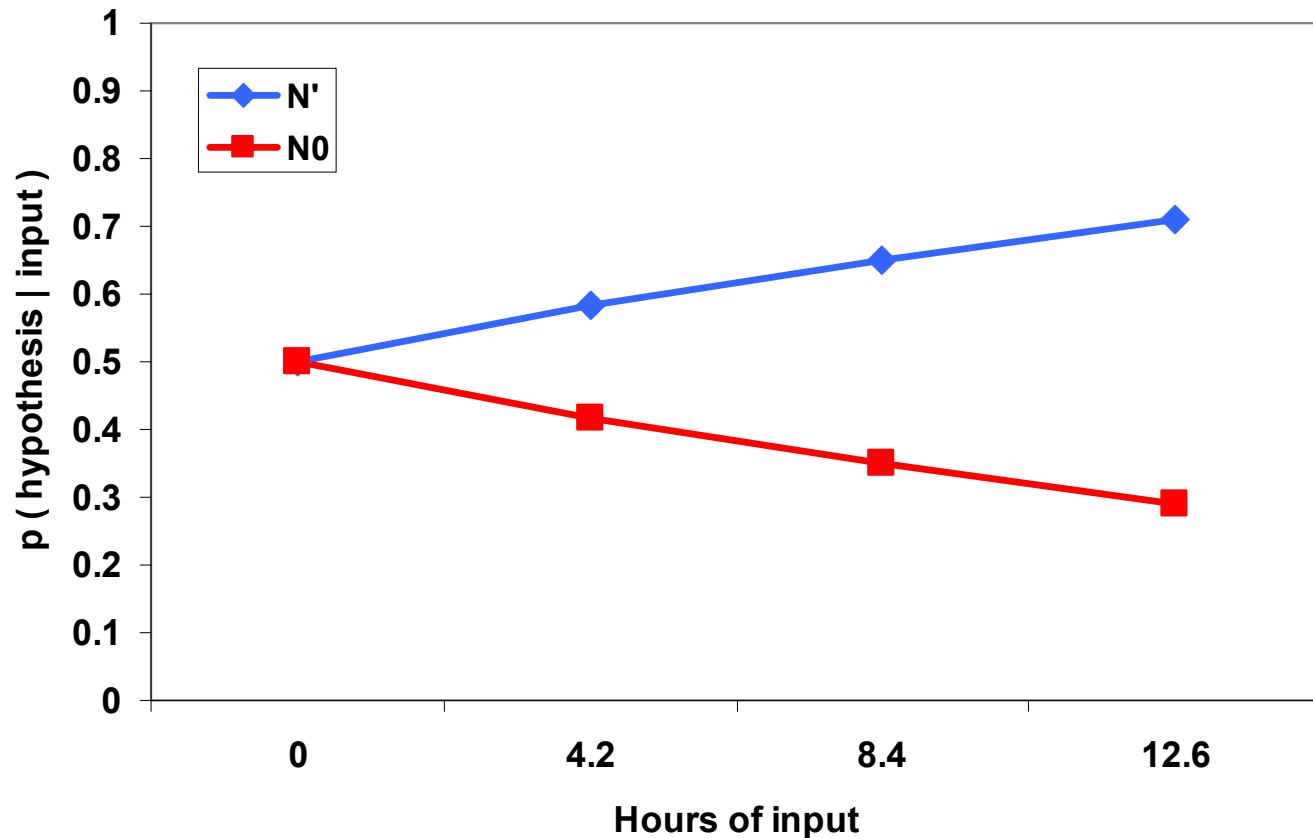
N' grammar **does not**.

So N' grammar should have **tighter data-fit**, and thus **higher likelihood**.

Results



Results



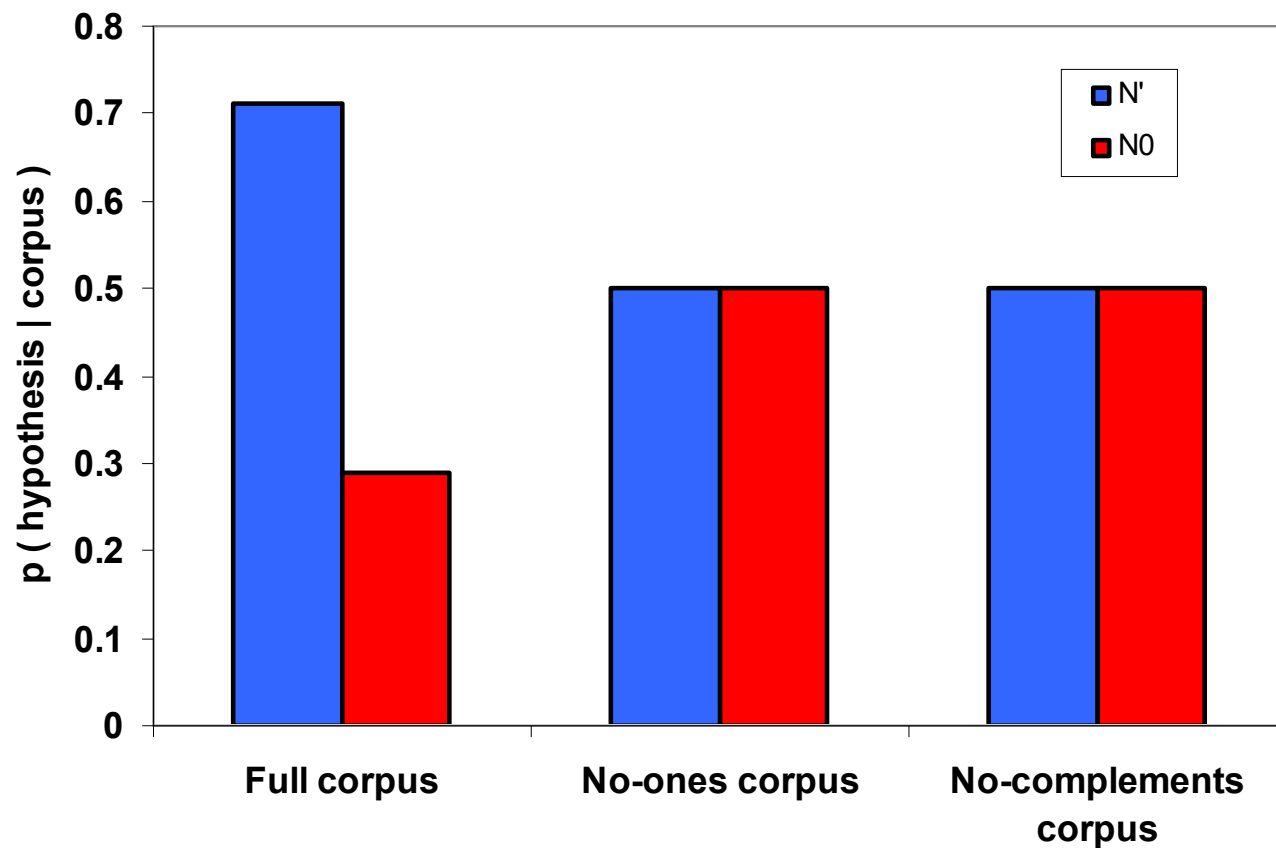
cf. “entrenchment”: Braine, Brooks, Tomasello, Theakston...

Targeted impoverishment of stimulus

Explore what **aspect** of the data is critical to learning.

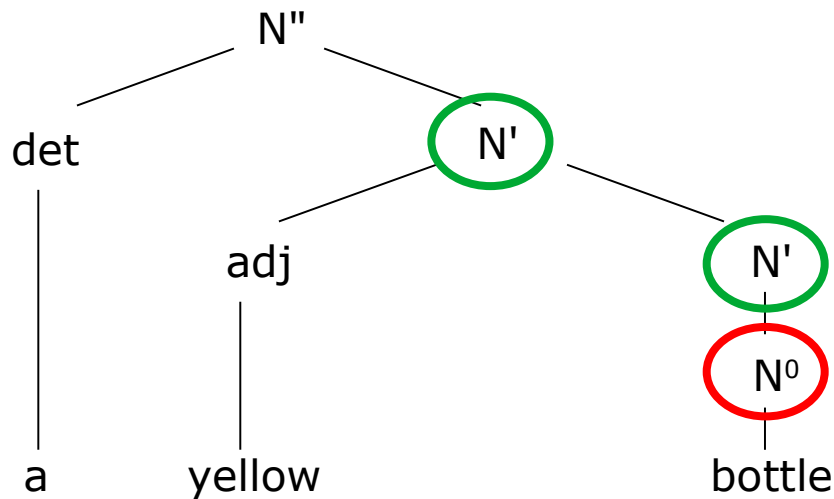
- No-ones (.6% excluded)
- No-complements (.8% excluded)

Results



In sum

Not necessarily innate, even though the relevant input is proportionally rare:



What **questions** do you have?

What **comments** do you have?

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- Foraker et al. on anaphoric *one*
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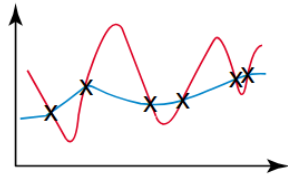
Simplicity: a unifying principle in cognitive science?

Nick Chater^a and Paul Vitányi^b

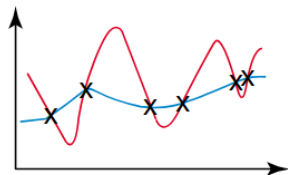
Core idea

Much of perception, learning and high-level cognition involves **finding patterns** in data. But there are always **infinitely** many patterns compatible with any finite amount of data. How does the cognitive system choose '**sensible**' patterns? A long tradition ... argues that patterns 'should' be chosen according to **how simply they explain the data**.

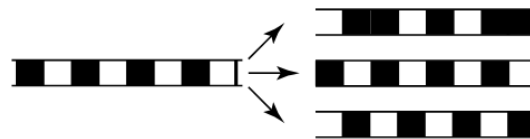
(a)



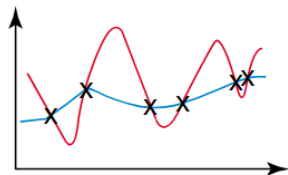
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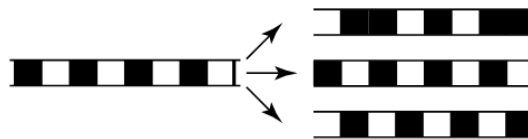
(b)



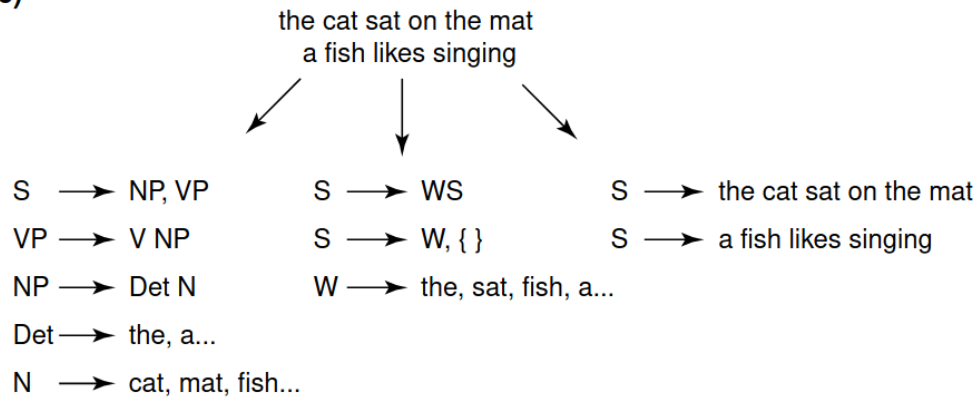
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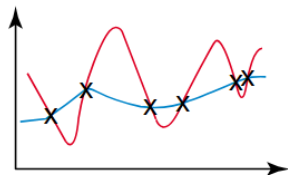
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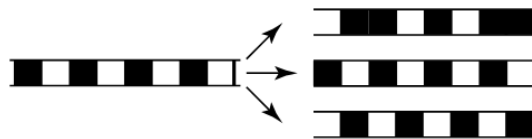
(c)



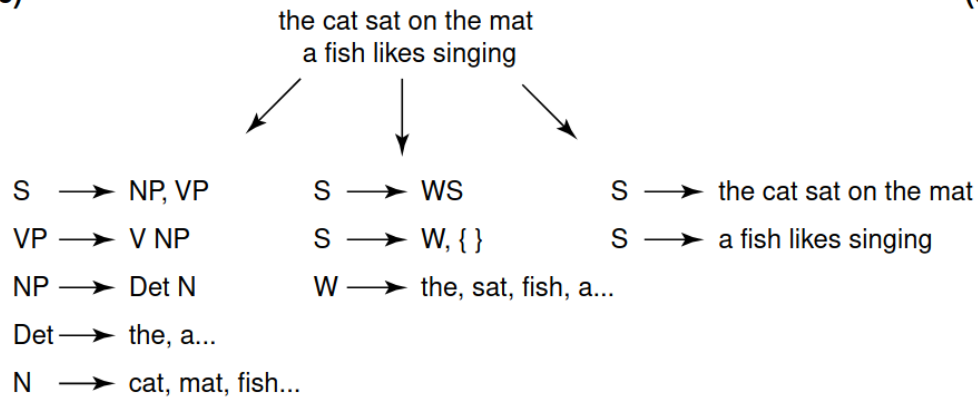
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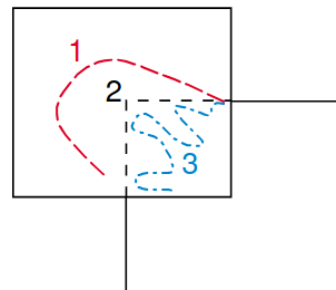
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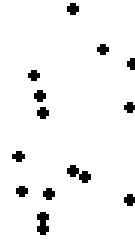
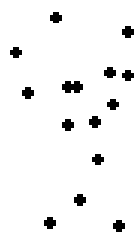


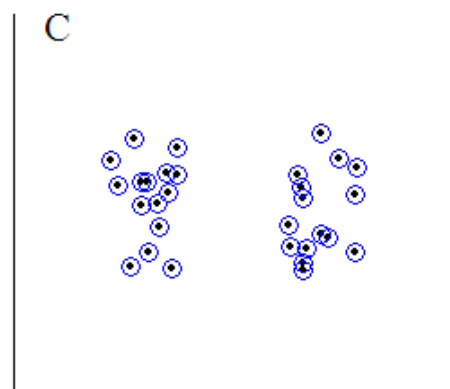
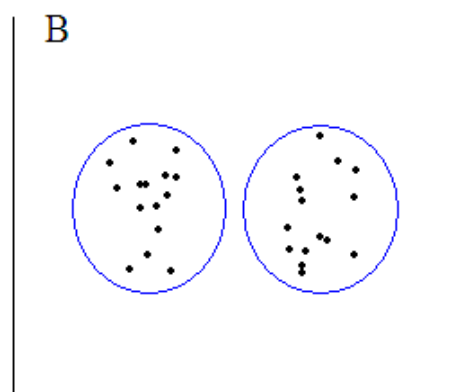
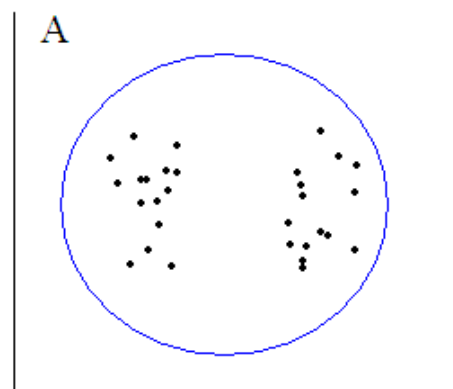
(c)



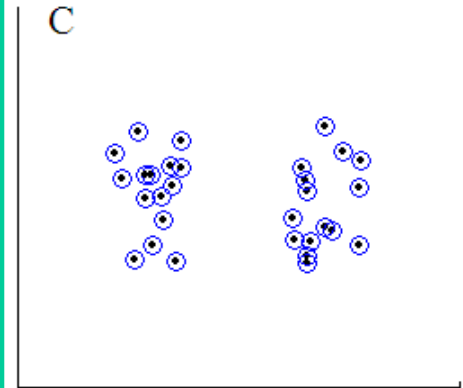
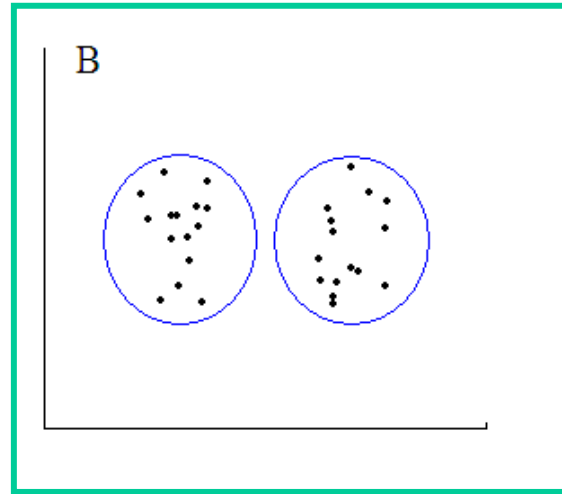
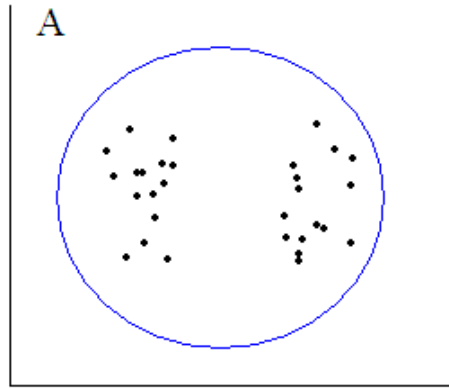
(d)







Optimizes **tradeoff** between
simplicity and data fit



Proposal:
Language is learned in a similar way

the cat sat on the mat
a fish likes singing



S $\rightarrow$ NP, VP

VP $\rightarrow$ V NP

NP $\rightarrow$ Det N

Det $\rightarrow$ the, a...

N $\rightarrow$ cat, mat, fish...

S $\rightarrow$ WS

S $\rightarrow$ W, { }

W $\rightarrow$ the, sat, fish, a...

S $\rightarrow$ the cat sat on the mat

S $\rightarrow$ a fish likes singing

the cat sat on the mat
a fish likes singing



S → NP, VP

VP → V NP

NP → Det N

Det → the, a...

N → cat, mat, fish...

S → WS

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W → the, sat, fish, a...

S → the cat sat on the mat

S → a fish likes singing



Perfect data fit,
but very complex

the cat sat on the mat
a fish likes singing



S → NP, VP

VP → V NP

NP → Det N

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N → cat, mat, fish...

S → WS

S → W, { }

W → the, sat, fish, a...

S → the cat sat on the mat

S → a fish likes singing



Very simple, but
poor data fit

the cat sat on the mat
a fish likes singing



S $\longrightarrow$ NP, VP

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Optimizes **tradeoff** between
simplicity and data fit
(e.g. Perfors et al. 2011)

What **questions** do you have?

What **comments** do you have?

This week's readings

- Chomsky, *Knowledge of language*
- Foraker et al. on anaphoric *one*
- Chater & Vitanyi on simplicity
- Pullum on Chomsky 2011 in London

- Polemic alert!

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- Rocks and kittens – general nativism vs. linguistic nativism.

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- Genetic fixity

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- Rocks and kittens – general nativism vs. linguistic nativism.
- **The lesson of Bayes' theorem**
- Genetic fixity

“Many linguists have given the impression that they think it is **impossible** to learn from **positive** data that something is **not grammatical**.

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Lightfoot (1998: 585) suggests, for example, that although you can perhaps learn from experience that **auxiliary reduction** is optional in the interior of a clause, **you cannot possibly learn that it is forbidden at the end of a clause**; hence linguistic nativism has to be true.

“Many linguists have given the impression that they think it is **impossible** to learn from **positive** data that something is **not grammatical**.

Lightfoot (1998: 585) suggests, for example, that although you can perhaps learn from experience that **auxiliary reduction** is optional in the interior of a clause, **you cannot possibly learn that it is forbidden at the end of a clause**; hence linguistic nativism has to be true.

This reasoning is flawed, and Bayes' Theorem teaches us why.”

Pullum vs. Lightfoot on auxiliary reduction

d1 = Pat is younger than Kim is.

d2 = Pat's younger than Kim is.

d3 = *Pat is younger than Kim's.

d4 = *Pat's younger than Kim's.

Pullum vs. Lightfoot on auxiliary reduction

- **H_o** : aux reduction is **optional** both clause-internally and clause-finally.
- **H_f** : aux reduction is optional clause-internally but **forbidden** clause-**finally**.

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	$p(\cdot H_o)$	$p(\cdot H_f)$
d1 = Pat is younger than Kim is.		
d2 = Pat's younger than Kim is.		
d3 = *Pat is younger than Kim's.		0
d4 = *Pat's younger than Kim's.		0

Pullum vs. Lightfoot on auxiliary reduction

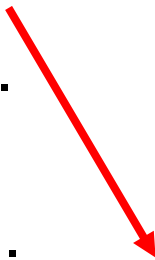
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“Wasted” probability mass!



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